

ProjectPitch-GliderRodeo

Join in Group Note-Taking & Discussions
[Project Pitch Discussions](#)

Glider Selection Dashboard

— Match requirements to configurations

- Slack channel: # glider-selection-dashboard
- GitHub repo: /glider-selection-dashboard



1. Define Mission Constraints

Users input environmental profiles and project parameters:

- Mission duration, geographic area & distance/area to cover
- Bathymetry, operating depth & environmental conditions
- Species/phenomena of interest & required sensors
- Sampling rate, duty cycle & real-time vs. archival
- Operational constraints, endurance, battery type & budget



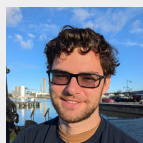
2. Intelligent Suitability Assessment

The system evaluates and optimizes platform configurations:

- Generates a ranked shortlist with distinct fit justifications
- Side-by-side platform comparison & trade-off analysis
- Pre-deployment risk identification for battery & telemetry limits
- Potential Data Sources/Tools: Manufacturer websites
- Potential Challenges: ??
- First steps: Where to create the Dashboard? R (shinydashboard)?

AI inspiration



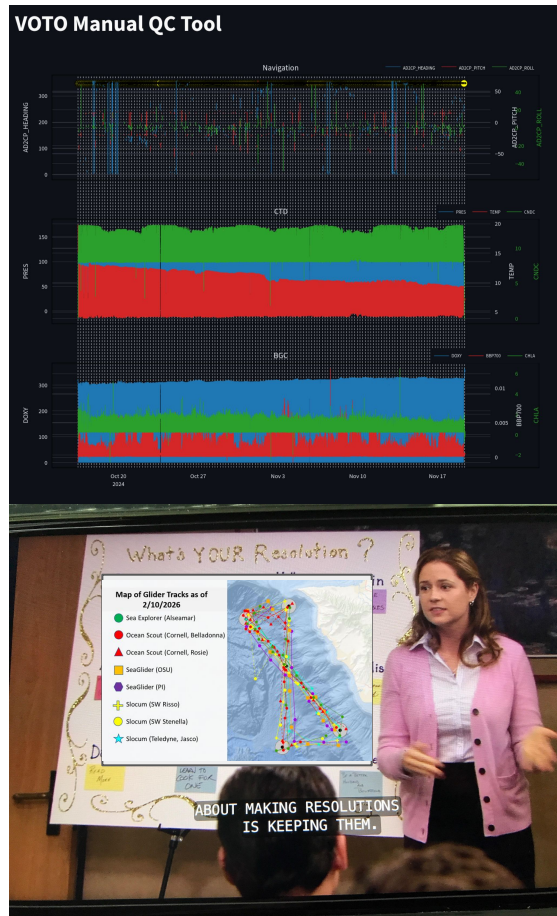


PAM data transform visualizer

- One sentence goal: Build a simple visualizer to see how transformational data operations change PAM data products.

name=pending

- Slack channel: # pam-visualizer
- GitHub repo: /pam-visualizer
- Team members: AJ, <hopefully someone else>
 - have experience with interactive plotting in Python
 - need experience with PAM data (where to find it and how is it structured), operations/common glider processing steps
- Potential Data Sources/Tools
 - Some raw data from authorized sources (NCEI), final data shared
- Potential Challenges
 - Testing on different systems
 - Large file sizes
- First steps?
 - Download the data, try to plot with **plotly**, **bokeh**, or **streamlit**



Noise Effect Calculator for MarMam Detection

Goal: Develop a tool that can be applied to acoustic data and describe the relationship between noise and marine mammal detectability.

Slack: #noise-fx-calc

GitHub: /noise-fx-calc

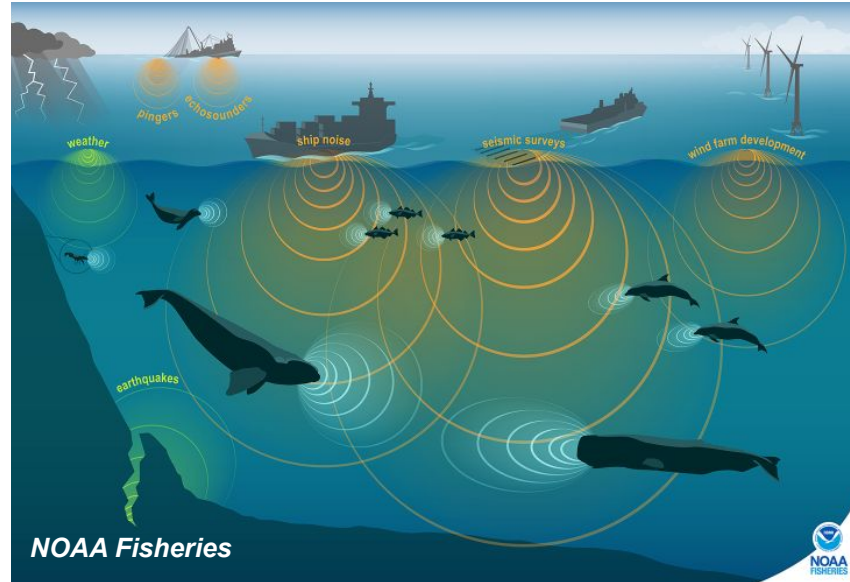
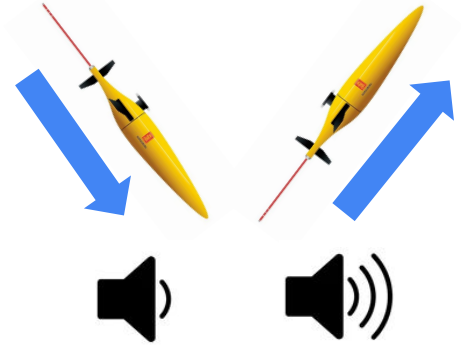
Team members: Will (Aaron) Deans... *All are welcome!*

- *Experienced with:* PAM data, bioacoustics, MATLAB
- ***Need experience with:*** Glider data/ops, Python, environmental data, statistics?

Potential data sources/tools: Band levels, environmental data, profile data, MarMam detections

Potential challenges: Deciding detection thresholds

First steps: Explore the data!



Sub-proj/second proj?? Noise Profiler

Goal: Develop a tool that can be applied to glider acoustic data and describe noise conditions as a function of glider mode, man-made noise, and environmental variables.

Slack: #noise-fx-calc

GitHub: /noise-fx-calc

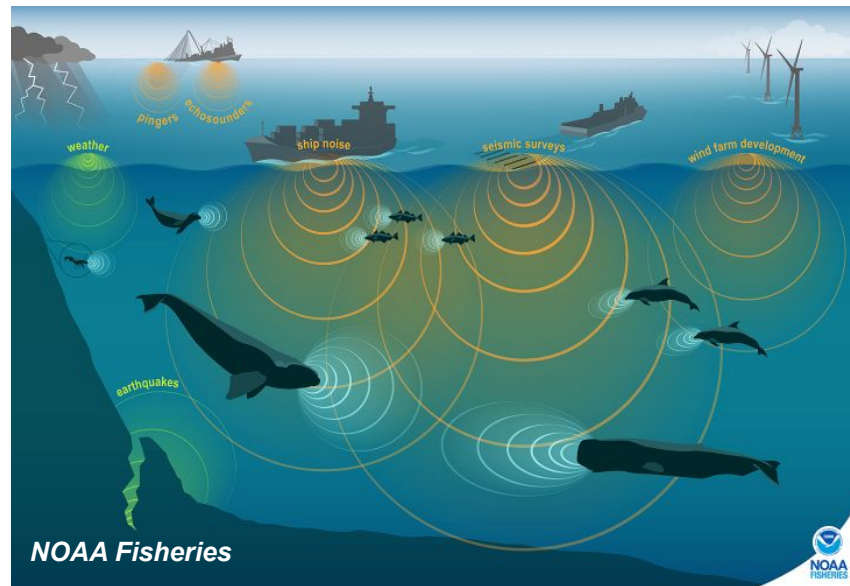
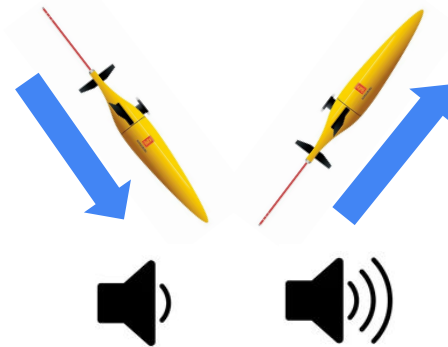
Team members: Will (Aaron) Deans... *All are welcome!*

- *Experienced with:* PAM data, bioacoustics, MATLAB
- ***Need experience with:*** Glider data/ops, Python, environmental data, statistics

Potential data sources/tools: Band levels, environmental data, profile data, *raw acoustic data?*

Potential challenges: Variable instrument sensitivity

First steps: Organize the data!



Glider Performance by Mission Modes

Slack: #glider-performance

GitHub: /glider-performance

We need to understand how gliders work in different situations to inform how we use them.

- How does battery endurance vary by Mission Mode?
- How noisy are they in different Mission Modes?
- etc.

Results could inform other projects

→ No Project Lead

→ Project? Discussion? Online or Offline?



1



Trimming

Fine-tune adjustments to prepare for rodeo!

2



FAST Mission Mode

Pilot to prioritize trackline progression

3



SLOW Mission Mode

Pilot to minimize noise while reaching end of line

4



INTERMEDIATE Mission Mode

Balance maintaining course w/ reduced self-noise

5



Drift Mode

Intermediate mode w/ 20 min drift @ 800m (repeated)

6



Mini Rodeo

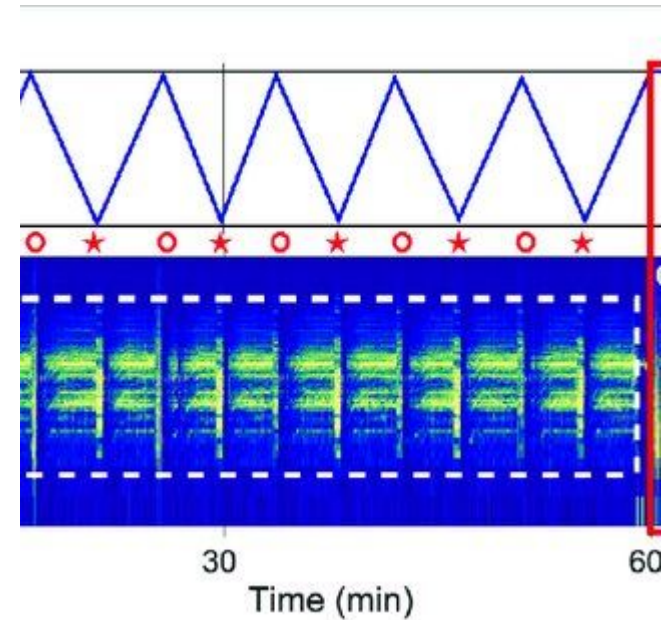
Cycle all gliders through Phase 2-5 (12 hrs each)

Glider Noise Catalog

Slack: #glider-noise

GitHub: /glider-noise

- What noises do they make?
- How do we document this?
 - Standard Annotations?
 - Link Pilot/Engineering to Audio?
- Where do we document this?
 - Where to store Audio clips?
 - Where to store Annotations?
- Other Ideas?
- Glider Rodeo Data → Conversation?
- Project Leader?



Sleepy Pilot Warning: Developing a warning system that evaluates environmental conditions, mission plans, and glider attitude to predict the chances pilots will have a sleepless night (and send a warning!)

- One sentence goal:
- Slack channel: #projects/sleepyPilots
- GitHub repo: <https://github.com/xxx/sleepyPilots>
- Team members: ABC, XYZ, IJK
 - have experience with ...
 - need experience with ...
- Potential Data Sources/Tools
- Potential Challenges
- First steps?

